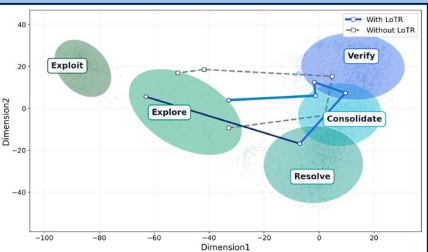


A conference organizer needs to assign one volunteer to the registration desk. There are three volunteers: Anna, Ben, and Cara. The desk requires someone who **has a laptop**, **can arrive before 8:30**, and **has not been assigned to the poster session**. *Anna* has a laptop and can arrive at 8:20, but she is assigned to the poster session. *Ben* has no laptop but can arrive at 8:10 and is not assigned to the poster session. *Cara* has a laptop, can arrive at 8:25, and is not assigned to the poster session. Who should be assigned to the registration desk, and why?



Chain-of-Thought with LoTR



	C1	C2	C3	C4	C5
①	0.02	0.18	0.07	0.68	0.05
②	0.03	0.64	0.18	0.10	0.05
③	0.03	0.14	0.65	0.05	0.13
④	0.04	0.12	0.63	0.04	0.17
⑤	0.08	0.10	0.22	0.03	0.57
⑥	0.70	0.04	0.08	0.02	0.16

- ① There are three volunteers and three requirements for the registration desk.
- ② Choice must satisfy all three conditions simultaneously: laptop, before 8:30, and not assigned to poster session.
- ③ *Anna* has a laptop and arrives early, but she is assigned to the poster session, so she is excluded.
- ④ *Ben* arrives early and is not assigned, but he does not have a laptop, so he is excluded.
- ⑤ *Cara* has a laptop, arrives at 8:25, and is not assigned to the poster session; after eliminating Anna and Ben, Cara is the only valid choice.
- ⑥ Therefore, **Cara** should be assigned.



Chain-of-Thought without LoTR

- ① The desk needs someone with a laptop and early arrival.
- ② *Anna* has a laptop and can arrive at 8:20, which is before 8:30.
- ③ *Ben* arrives earlier, but he has no laptop.
- ④ Since *Anna* satisfies the main desk requirements, she is the best choice. (pathway mismatch)
- ⑤ Therefore, **Anna** should be assigned.

